1.4.2 Electric field due to a dipole

Case (i) Electric field due to an electric dipole at points on the axial line

Consider an electric dipole placed on the x-axis as shown in Figure 1.15. A point C is located at a distance of r from the midpoint O of the dipole on the axial line.

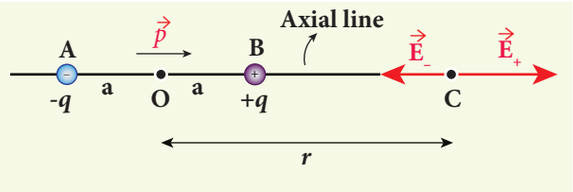


Figure 1.15 Electric field of the dipole along the axial line

The electric field at a point C due to $+q$ is

$$\vec{E}_+ = \frac{1}{4\pi\epsilon_0}\frac{q}{(r-a)^2} \text{ along BC}$$

Since the electric dipole moment vector $\vec{p}$ is from $-q$ to $+q$ and is directed along BC, the above equation is rewritten as

$$\vec{E}_+ = \frac{1}{4\pi\epsilon_0}\frac{q}{(r-a)^2}\hat{p} \qquad (1.13)$$

where $\hat{p}$ is the electric dipole moment unit vector from $-q$ to $+q$.

The electric field at a point C due to $-q$ is

$$\vec{E}_- = -\frac{1}{4\pi\epsilon_0}\frac{q}{(r+a)^2}\hat{p} \qquad (1.14)$$

Since $+q$ is located closer to the point C than $-q$, $\vec{E}_+$ is stronger than $\vec{E}_-$. Therefore, the length of the $\vec{E}_+$ vector is drawn larger than that of $\vec{E}_-$ vector.

The total electric field at point C is calculated using the superposition principle of the electric field.

$$\vec{E}_{tot} = \vec{E}_+ + \vec{E}_-$$
$$= \frac{1}{4\pi\epsilon_0}\frac{q}{(r-a)^2}\hat{p} - \frac{1}{4\pi\epsilon_0}\frac{q}{(r+a)^2}\hat{p}$$

$$\vec{E}_{tot} = \frac{q}{4\pi\epsilon_0}\left(\frac{1}{(r-a)^2} - \frac{1}{(r+a)^2}\right)\hat{p} \qquad (1.15)$$

$$\vec{E}_{tot} = \frac{1}{4\pi\epsilon_0}q\left(\frac{4ra}{(r^2-a^2)^2}\right)\hat{p} \qquad (1.16)$$

Note that the total electric field is along $\vec{E}_+$, since $+q$ is closer to C than $-q$. The direction of $\vec{E}_{tot}$ is shown in Figure 1.16.

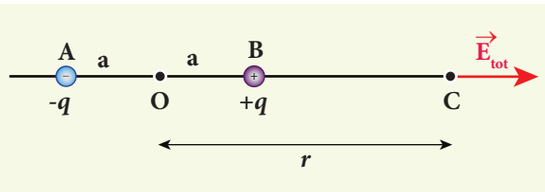


Figure 1.16 Total electric field of the dipole on the axial line

If the point C is very far away from the dipole ($r >> a$). Then under this limit the term $(r^2-a^2)^2 \approx r^4$. Substituting this into equation (1.16), we get

$$\vec{E}_{tot} = \frac{1}{4\pi\epsilon_0}\left(\frac{4aq}{r^3}\right)\hat{p} \quad (r >> a)$$

since $2aq\,\hat{p} = \vec{p}$

$$\vec{E}_{tot} = \frac{1}{4\pi\epsilon_0}\frac{2\vec{p}}{r^3} \quad (r >> a) \qquad (1.17)$$

If the point C is chosen on the left side of the dipole, the total electric field is still in the direction of $\vec{p}$. We infer this result by examining the electric field lines of the dipole shown in Figure 1.14(b).

Case (ii) Electric field due to an electric dipole at a point on the equatorial plane

Consider a point C at a distance r from the midpoint O of the dipole on the equatorial plane as shown in Figure 1.17. Since the point C is equi-distant from $+q$ and $-q$, the magnitude of the electric fields at C due to $+q$ and $-q$ are the same. The direction

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